	PC		Project Management	3	0	2	5	4
	PC		Earthquake Engineering	3	1	0	4	4
	Pro		**Seminar**					1
				18	**2**	**2**		**24**
Semester 8	**Pro**		Major/Comprehensive Project					**12**

Teaching Scheme					Major Project					
					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
			12	--	--	--	--	25	25	50

COURSE OBJECTIVES

1. To demonstrate expertise in the chosen project topic, showcasing robust technical knowledge.
2. To apply engineering principles to address real-world challenges faced by industry and society.
3. To develop proficiency in solving intricate problems using a systematic approach.
4. To communicate effectively, both in written and oral formats, with industry stakeholders and the broader community.

Details:

Students enrolled in the Civil Engineering program are required to select a research topic within the realm of current practices in the field. Collaborative project work is encouraged, with group sizes limited to a maximum of three students. Research areas encompass a wide spectrum including Structural Design and Analysis, Soil and Geotechnical Engineering, Traffic Engineering, Pavement Design and Analysis, Metro and Monorail Construction, Airport Planning and Design, Water Resource Management with GIS and Surveying, Construction Techniques and Management, as well as Environmental Engineering and Energy Conservation.

Each group must select a faculty member from the department to serve as their guide. The chosen research topic should align with the interests and expertise of both the group and the guide. Additionally, students undertaking major projects are required to include a one-page section outlining the novelty of their work.

Upon completion, the thesis must be formatted according to PDPU standards and submitted to the department through the designated guide. One copy, signed by both the guide and the Head of the Department, will be archived in the departmental library for record-keeping and to verify originality. A review committee consisting of at least two faculty members will assess the internal components of the project and dissertation work, while the external review panel will include an external examiner, the department head, and the guide.

COURSE OUTCOMES

CO-1: Define the relevance of the project topic utilizing established techniques in Civil Engineering.

CO-2: Summarize the problem statement with analytical and documentation skills, informed by literature surveys.

CO-3: Apply gathered data to formulate project plans.

CO-4: Solve problems using contemporary tools and techniques, supported by critical investigation.

CO-5: Prepare a comprehensive project report adhering to institutional guidelines.

CO-6: Present project findings effectively using accepted tools like PowerPoint.

					Comprehensive Project					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
			12	--	--	--	--	25	25	50

COURSE OBJECTIVES

1. To understand civil engineers' roles on construction sites.
2. To grasp contract processes and governmental policies.
3. To master structural design and analysis for diverse projects.
4. To develop skills in scheduling and work execution management.

Details:

Students and the university are tasked with selecting a company specializing in Civil Engineering for an individual comprehensive project. The project scope encompasses various areas such as Structural Design and Analysis, Soil and Geotechnical Engineering, Traffic Engineering, Pavement Design and Analysis, Metro and Monorail Construction, Airport Planning and Design, Water Resource Management with GIS and Surveying, Construction Techniques and Management, as well as Environmental Engineering and Energy Conservation.

Each student must choose a mentor from the department and a supervisor from the industry. The selected project area should be mutually agreeable to both the student and the mentor.

Upon completion, the project report must be formatted according to PDPU standards and submitted to the department through the industry supervisor and departmental mentor, accompanied by an industry training certificate. One copy of the thesis, endorsed by the mentor and the Head of the Department, will be archived in the departmental library to ensure originality and record-keeping.

A review committee comprising a minimum of two faculty members will evaluate the internal components of the comprehensive project work, while the external review panel will consist of an external examiner, the department head, and the mentor.

COURSE OUTCOMES

CO-1: Define the relevance of comprehensive project work area selected for the study.

CO-2: Understand the role and responsibility of a civil engineering at construction site.

CO-3: Analyse the various civil engineering components as per site requirement.

CO-4: Apply the knowledge in execution of work in a systematic manner

CO-5: Prepare schedule of workers, material and equipment requirement for day to day work execution.

CO-6: Practice the acquired knowledge, skills and attitudes for becoming a professional engineer